Lab 2: Exploratory Data Analysis

NAME:

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NAME:

0.1 Setup

```
library("knitr")
library("dslabs")
library("dplyr")
library("ggplot2")

# Optional packages used later:
#install.packages("Hmisc")
#install.packages("gtsummary")
#install.packages("GGally")

diseases = us_contagious_diseases

knitr::opts_chunk$set(fig.align = "center")
```

1 Exploring the Variables

One of the first things we should do after loading in a dataset is find out what are the variables. Do you remember the function to do this?

```
names(diseases)
```

```
[1] "disease"         "state"           "year"            "weeks_reporting"
[5] "count"           "population"
```

```
str(diseases)
```

```
'data.frame':   16065 obs. of  6 variables:
 $ disease        : Factor w/ 7 levels "Hepatitis A",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ state          : Factor w/ 51 levels "Alabama","Alaska",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ year           : num  1966 1967 1968 1969 1970 ...
 $ weeks_reporting: num  50 49 52 49 51 51 45 45 45 46 ...
 $ count          : num  321 291 314 380 413 378 342 467 244 286 ...
 $ population     : num  3345787 3364130 3386068 3412450 3444165 ...
```

Next, we want to check how much data is missing. One way to do this is to use the command complete.cases:

```r
sum(!complete.cases(diseases))
```

```
[1] 214
```

```r
nrow(diseases)
```

```
[1] 16065
```

This calculates the number of incomplete rows (some missing data) and then compares it to the total number of rows in the dataset. Since we are not missing much data in this case, we can choose to omit these rows using the na.omit command:

```r
diseases = na.omit(diseases)
```

Otherwise, we could explore which columns contain the most missing data. If a column contains a lot of missing values, it might indicate that we should ignore it. There are no such columns in our case. We will talk about more strategies for dealing with missing data later on in the course.

As mentioned in class, the **describe** function (which is part of the Hmisc package) can be very useful to get a first look at the columns in the dataset.

```r
library("Hmisc")
describe(diseases$year)
```

```
diseases$year
       n  missing distinct     Info     Mean  pMedian      Gmd      .05
   15851        0       84        1     1971     1972    26.33     1933
     .10      .25      .50      .75      .90      .95
    1938     1950     1975     1991     2000     2004

lowest : 1928 1929 1930 1931 1932, highest: 2007 2008 2009 2010 2011
```

```r
describe(diseases$disease)
```

```
diseases$disease
       n  missing distinct
   15851        0        7
```

```
Value      Hepatitis A     Measles       Mumps   Pertussis       Polio
Frequency         2346        3761        1785        2820        2027
Proportion       0.148       0.237       0.113       0.178       0.128

Value          Rubella    Smallpox
Frequency         1887        1225
Proportion       0.119       0.077
```

Note that describe doesn't work well for categorical variables with many levels such as state - we can always use summary instead.

We also might measure the variability of continuous variables by measuring their standard deviation:

```r
sd(diseases$count)
```

```
[1] 5921.36
```

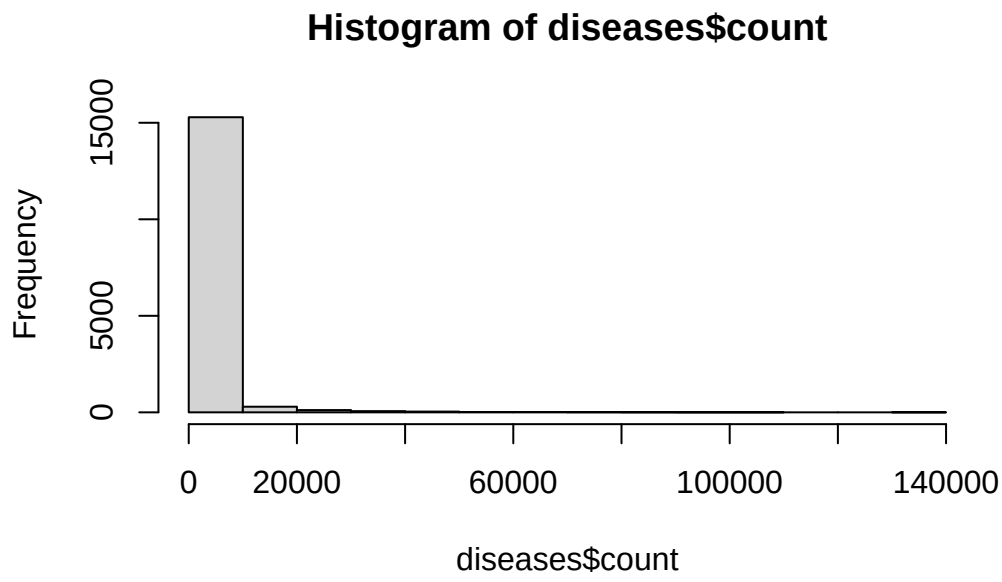
```r
mean(diseases$count)
```

```
[1] 1510.354
```

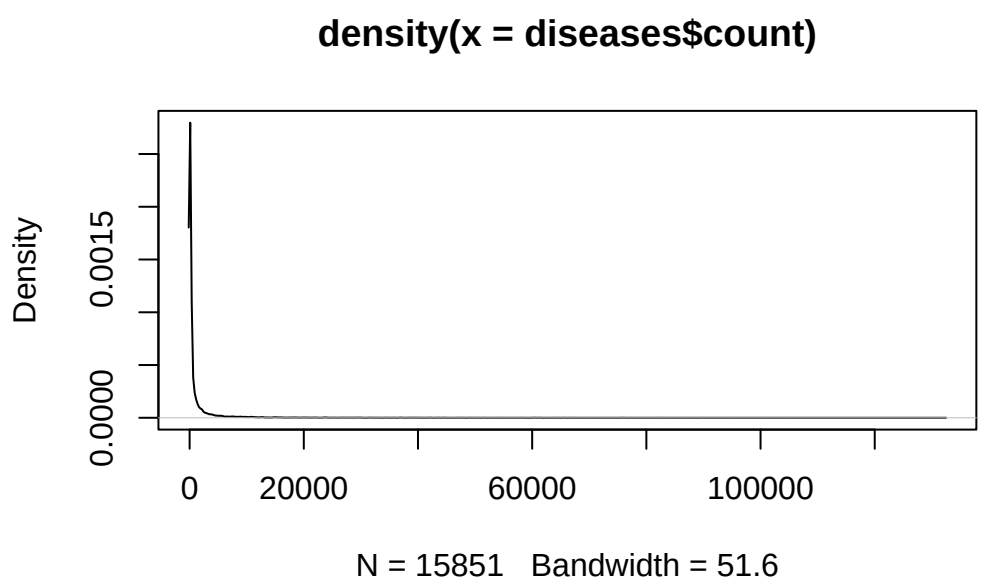
What do you notice about the standard deviation of count compared to the mean? Why might this be the case?

Beyond quantitative outputs, plotting the distribution of continuous variables can also be helpful.

```r
hist(diseases$count)
```

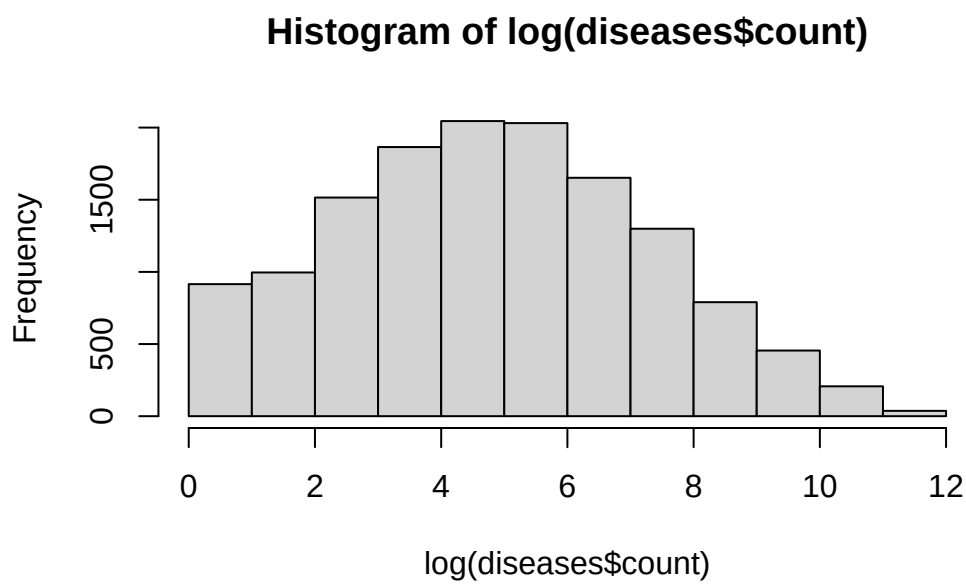



```r
plot(density(diseases$count))
```

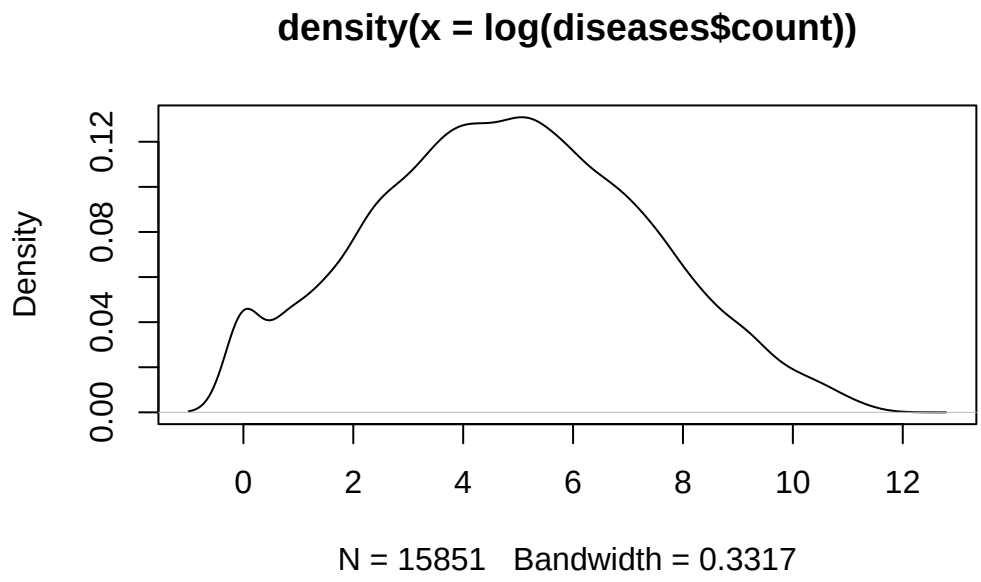


Of course, these plots are not very informative for count. Instead, let's look at a log transformation of the variable.

```
hist(log(diseases$count))
```

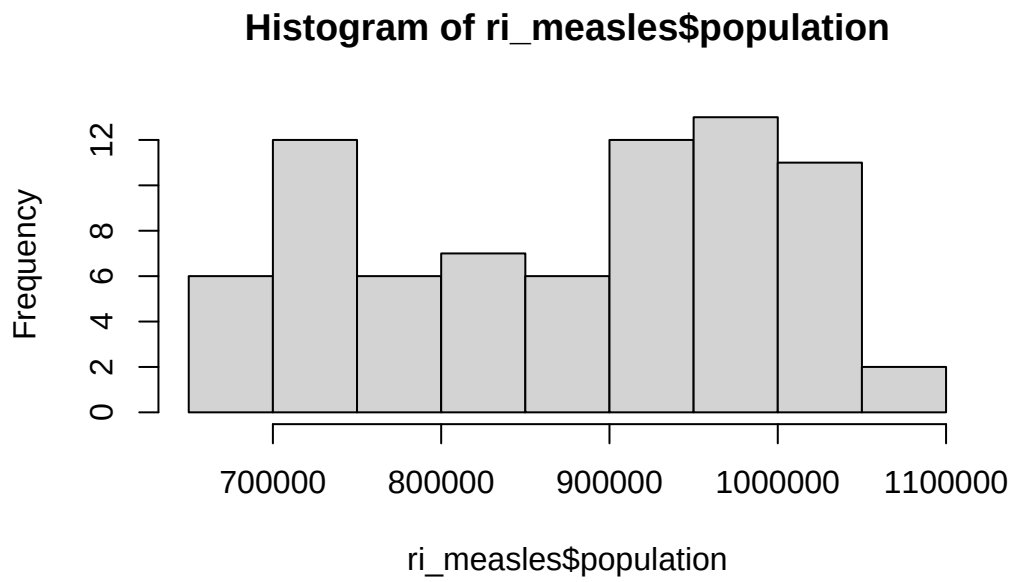



```
plot(density(log(diseases$count)))
```

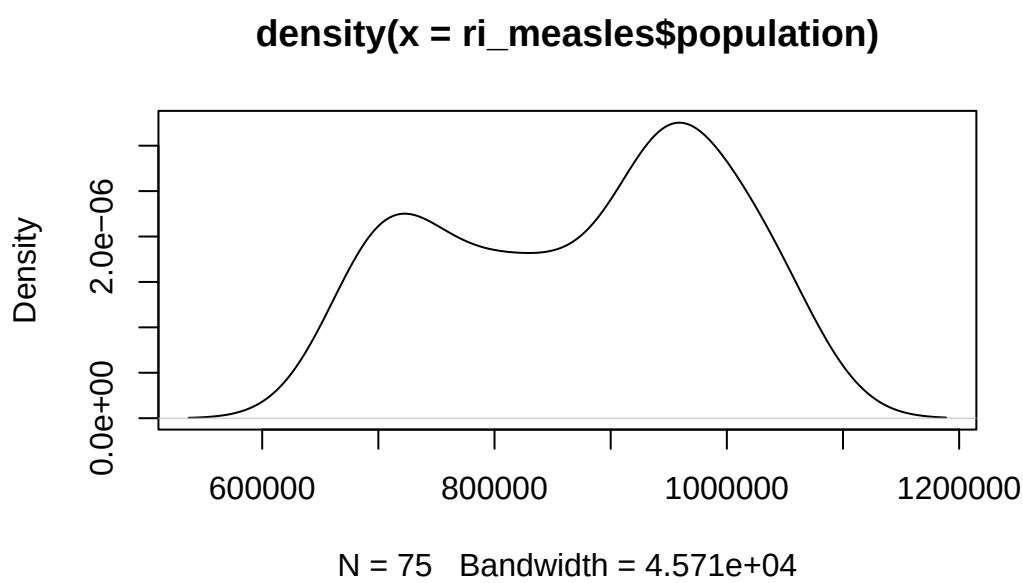


We can also try filtering the data and looking at another variable.

```
ri_measles = diseases %>%
  filter(state == "Rhode Island", disease == "Measles")

hist(ri_measles$population)
```

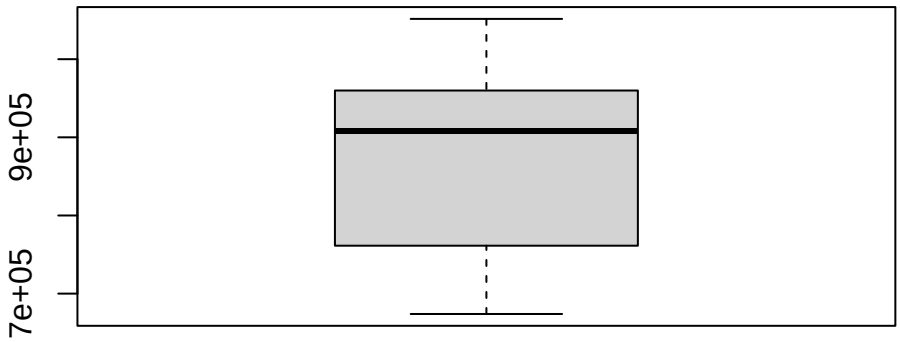



```
plot(density(ri_measles$population))
```



A box plot is also useful to help understand a distribution. Box plots can be especially useful for finding outliers and summarizing the variability of a variable.

```
boxplot(ri_measles$population)
```



2 Create rate

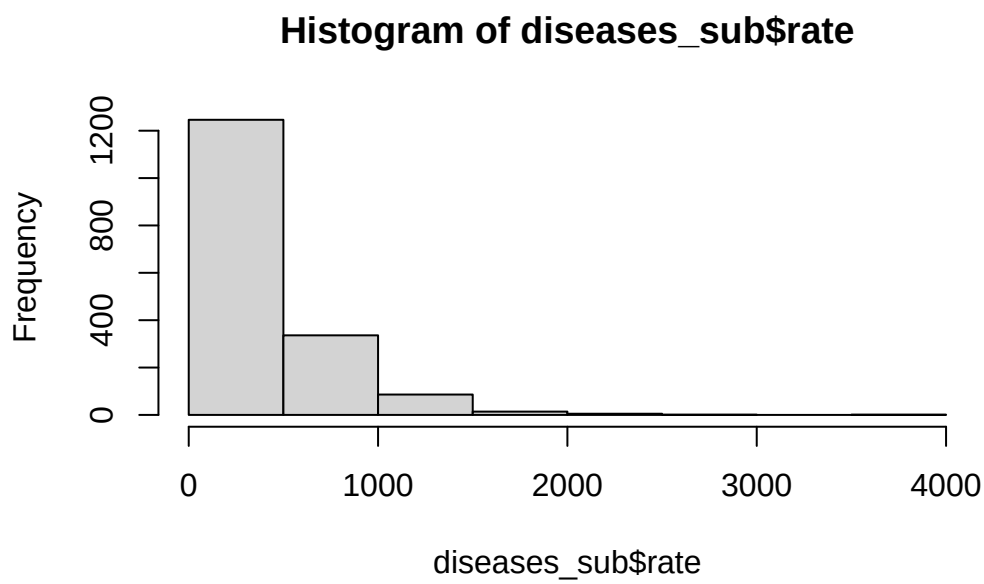
Create a new variable called rate which is equal to: 100000 × (count/population) × (52/weeks_reporting)

What does rate represent? Why might this new variable be helpful when looking at the distributions for different diseases?

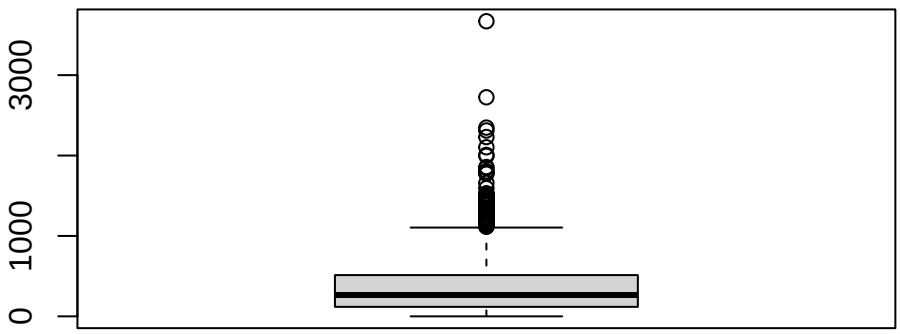
```
diseases = diseases %>%
  mutate(rate = 100000 * (count / population) * (52 / weeks_reporting))

# Remove cases where rate becomes NA/Inf (e.g., weeks_reporting = 0)
diseases = diseases %>%
  filter(is.finite(rate))
```

Looking only at records for measles and years prior to 1963 (the year the measles vaccine was introduced), create a histogram and boxplot for rate. Save this filtered dataset as **diseases_sub** for future use.

```
diseases_sub = diseases %>%
  filter(disease == "Measles", year < 1963)

hist(diseases_sub$rate)
```




```
boxplot(diseases_sub$rate)
```



3 Summary Tables (Basic vs Stratified)

3.1 Basic Summary Table

The basic summary table provides an overall description of the dataset without conditioning on any subgroup. It summarizes the central tendency and spread of key continuous variables such as year, weeks_reporting, count, population, and rate, and shows the frequency of each disease. This table is useful for gaining a high-level understanding of the scale of the data, the time span covered, and the typical magnitude of disease counts and rates across all observations. However, because it aggregates over all diseases and states, it can mask important heterogeneity between groups.

```r
summary(diseases[, c("year", "weeks_reporting", "count", "population", "rate")])
```

```
      year      weeks_reporting     count          population
 Min.   :1928   Min.   : 1.00   Min.   :     0   Min.   :   86853
 1st Qu.:1948   1st Qu.:38.00   1st Qu.:    17   1st Qu.: 1089254
 Median :1973   Median :47.00   Median :   110   Median : 2827642
 Mean   :1970   Mean   :42.11   Mean   :  1683   Mean   : 4254904
 3rd Qu.:1989   3rd Qu.:51.00   3rd Qu.:   677   3rd Qu.: 5162987
 Max.   :2011   Max.   :52.00   Max.   :132342   Max.   :37607525
```

```
      rate
 Min.   :   0.000
 1st Qu.:   1.098
 Median :   5.305
 Mean   :  61.909
 3rd Qu.:  29.809
 Max.   :3670.243
```

```r
table(diseases$disease)
```

```
Hepatitis A     Measles       Mumps   Pertussis       Polio     Rubella
       2327        3319        1576        2709        1844        1374
   Smallpox
       1079
```

or a nicer alternative:

```r
library("gtsummary")

diseases %>%
  select(year, weeks_reporting, count, population, rate, disease, state) %>%
  tbl_summary()
```

3.2 Stratified Summary Table (by disease)

The stratified summary table breaks down the same set of variables by disease, allowing us to compare distributions across different diseases. This makes it easier to identify differences in reporting frequency, population exposure, and disease severity. For example, we can observe whether certain diseases tend to have higher average rates or more variability than others. Stratification by disease is particularly important in this dataset because diseases differ substantially in prevalence, transmission patterns, and historical context.

```r
diseases %>%
  select(year, weeks_reporting, count, population, rate, disease) %>%
  tbl_summary(by = disease)
```

Characteristic	N = 14,228[1]
year	1,973 (1,948, 1,989)
weeks_reporting	47 (38, 51)
count	110 (17, 677)
population	2,827,642 (1,089,254, 5,162,987)
rate	5 (1, 30)
disease	
Hepatitis A	2,327 (16%)
Measles	3,319 (23%)
Mumps	1,576 (11%)
Pertussis	2,709 (19%)
Polio	1,844 (13%)
Rubella	1,374 (9.7%)
Smallpox	1,079 (7.6%)
state	
Alabama	286 (2.0%)
Alaska	168 (1.2%)
Arizona	294 (2.1%)
Arkansas	288 (2.0%)
California	312 (2.2%)
Colorado	295 (2.1%)
Connecticut	294 (2.1%)
Delaware	263 (1.8%)
District Of Columbia	251 (1.8%)
Florida	305 (2.1%)
Georgia	288 (2.0%)
Hawaii	194 (1.4%)
Idaho	293 (2.1%)
Illinois	306 (2.2%)
Indiana	290 (2.0%)
Iowa	288 (2.0%)
Kansas	282 (2.0%)
Kentucky	279 (2.0%)
Louisiana	286 (2.0%)
Maine	260 (1.8%)
Maryland	298 (2.1%)
Massachusetts	302 (2.1%)
Michigan	306 (2.2%)
Minnesota	290 (2.0%)
Mississippi	261 (1.8%)
Missouri	302 (2.1%)
Montana	269 (1.9%)
Nebraska	268 (1.9%)
Nevada	245 (1.7%)
New Hampshire	277 (1.9%)
New Jersey	298 (2.1%)
New Mexico	262 (1.8%)
New York	293 (2.1%)
North Carolina	285 (2.0%)
North Dakota	250 (1.8%)
Ohio	298 (2.1%)

Characteristic	Hepatitis A N = 2,327[1]	Measles N = 3,319[1]	Mumps N =
year	1,988 (1,977, 2,000)	1,962 (1,945, 1,978)	1,984 (1,976
weeks_reporting	41 (33, 48)	48 (41, 50)	48 (43,
count	142 (40, 422)	1,039 (65, 5,146)	48 (10, 3
population	3,342,087 (1,250,411, 5,804,815)	2,674,701 (993,534, 4,820,413)	3,514,889 (1,247,6
rate	7 (2, 17)	67 (4, 305)	2 (0, 1

[1]Median (Q1, Q3)

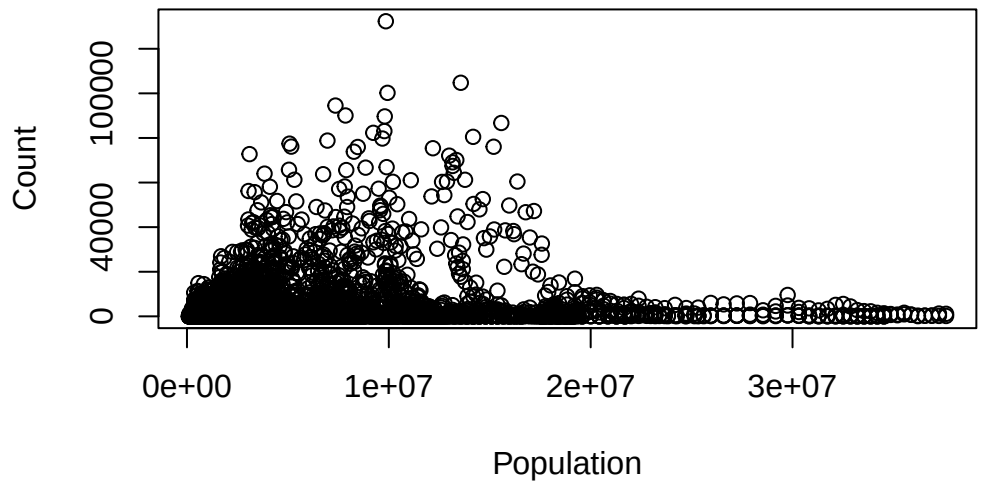
3.3 Why use both basic and stratified tables?

Using both basic and stratified summary tables allows us to balance **simplicity and insight**. The basic summary gives a concise overview of the entire dataset, while stratified summaries reveal patterns that depend on disease type or geographic location. Together, these tables help ensure that important subgroup differences are not overlooked during exploratory data analysis.

4 Relationships Between Variables

One thing that we noticed earlier is that the count is related to the total population size. A quick way to examine the relationship between two continuous variables is to plot them or calculate the correlation.

```
plot(x = diseases$population, y = diseases$count,
     xlab = "Population", ylab = "Count")
```

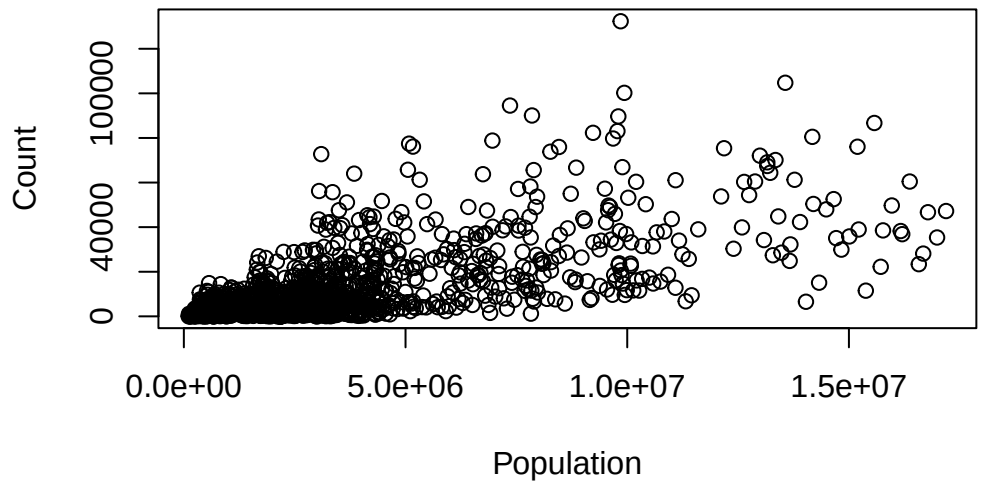



```
cor(diseases$count, diseases$population)
```

```
[1] 0.1240858
```

Do the results match your intuition? What if we do the same for diseases_sub?

```
plot(x = diseases_sub$population, y = diseases_sub$count,
     xlab = "Population", ylab = "Count")
```

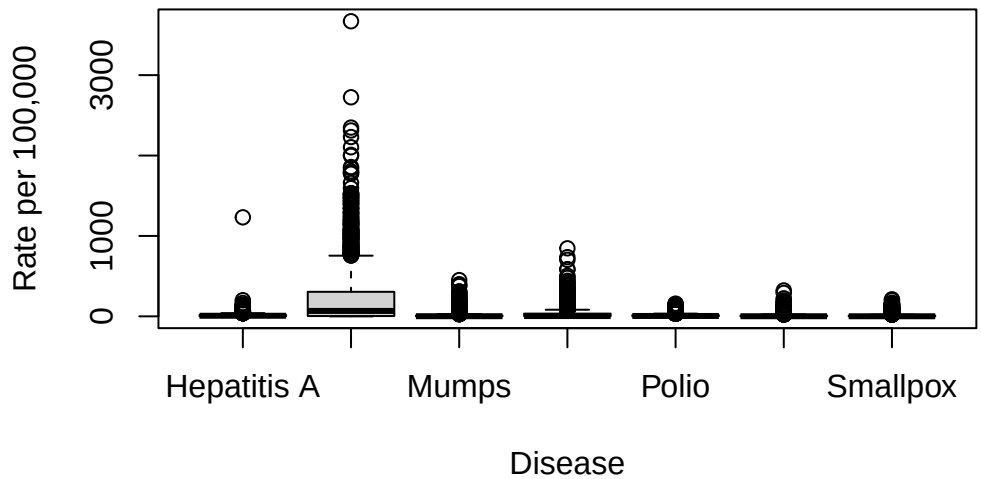



```
cor(diseases_sub$count, diseases_sub$population)
```

```
[1] 0.6903701
```

When we have one continuous variable and one categorical variable, then we can still use plot to compare the box plots for the continuous variable within each category. Try out the example below - you may need to zoom in to see the different box plots.

```
plot(x = diseases$disease, y = diseases$rate,
     xlab = "Disease", ylab = "Rate per 100,000")
```



What do you notice?

Last, when looking at the relationship between two categorical variables, we can use **contingency tables.**

First, create a new variable in the original diseases dataset called high_rate which represents whether or not the rate is above 100. Then, create a contingency table between high_rate and disease.

```
diseases = diseases %>%
  mutate(high_rate = (rate > 100))

table(diseases$high_rate, diseases$disease)
```

```
       Hepatitis A Measles Mumps Pertussis Polio Rubella Smallpox
  FALSE        2318    1843  1509      2425  1831    1357     1047
  TRUE            9    1476    67       284    13      17       32
```

Does this match with what we observed in the previous plot?

5 Data Visualization Tools

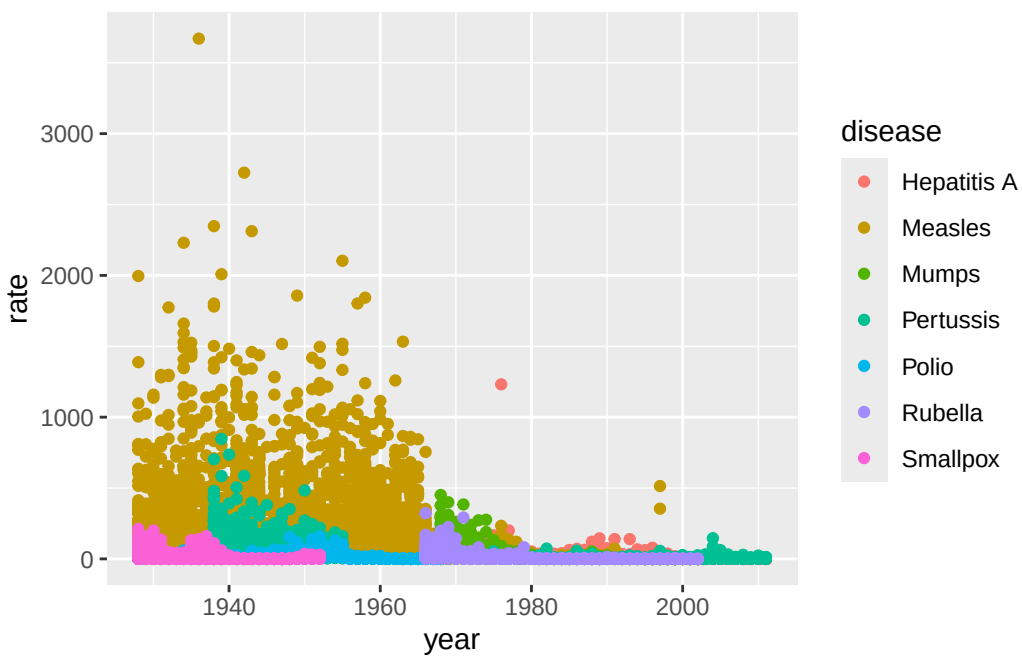
So far, we have just used built-in plotting functions in R. However, we can get much more creative by using the **ggplot2** package.

When using ggplot, we can add multiple pieces to a plot. We first start with the data.

```
p = ggplot(data = diseases)
```

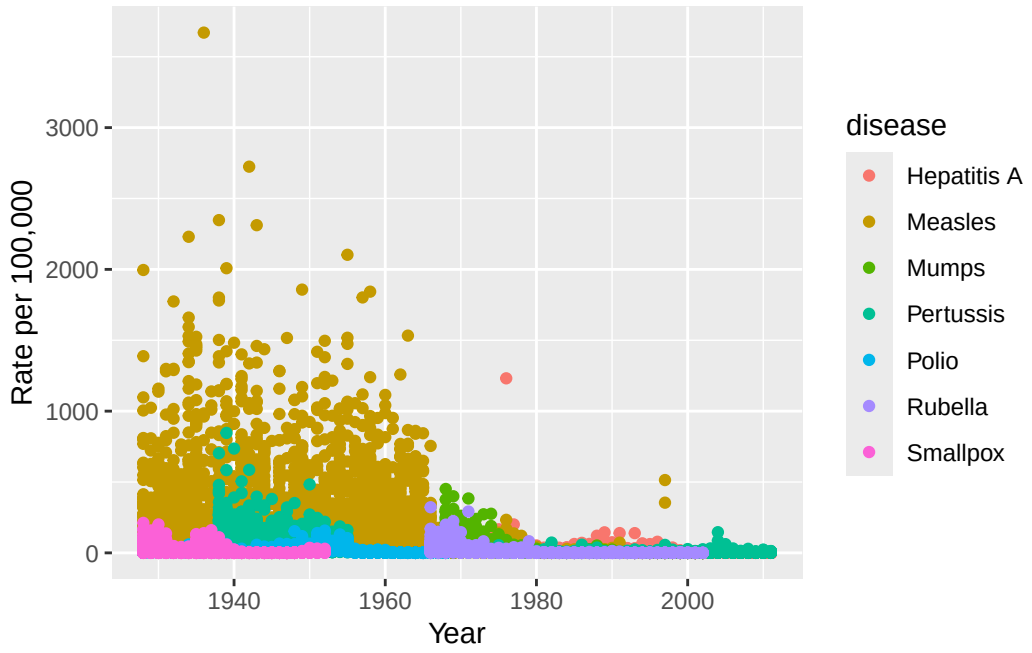
From this we can add on different **geoms**, which are layers. One such geom is a scatter plot.

```
p = p + geom_point(aes(x = year, y = rate, col = disease))
p
```



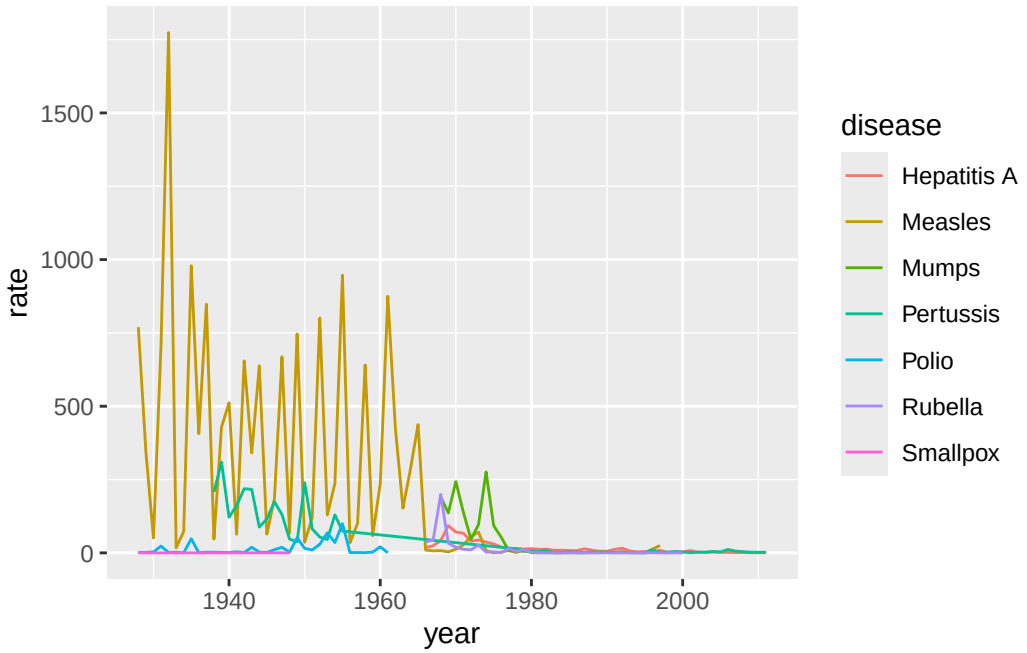
We can also add labels using the same framework.

```
p = p + xlab("Year") + ylab("Rate per 100,000")
p
```



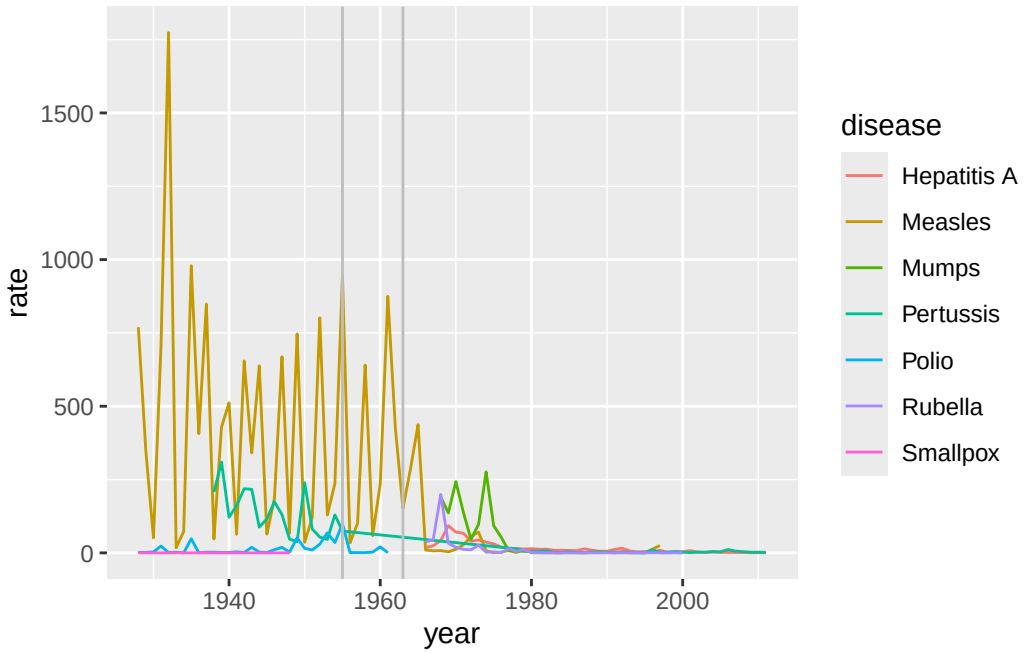
Let's try that again. This time, let's plot the overall R.I. rate for each disease over time.

```
p2 = diseases %>%
  filter(state == "Rhode Island") %>%
  ggplot() +
  geom_line(aes(x = year, y = rate, group = disease, col = disease))

p2
```



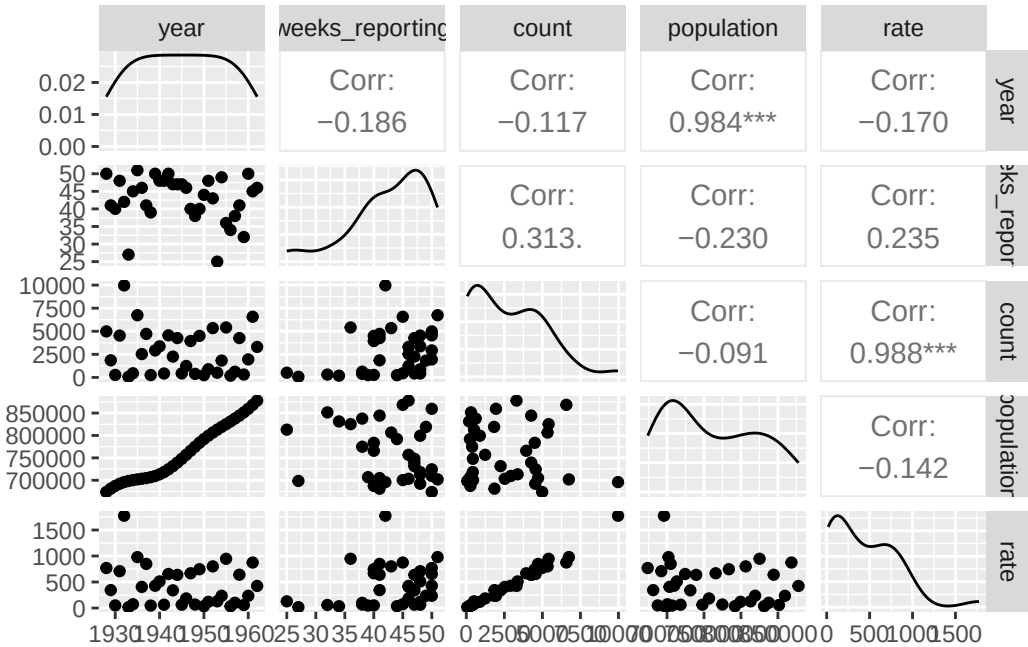
The measles vaccine was introduced in 1963 and the polio vaccine in 1955. Let's add vertical lines for each of these dates to highlight the effect of the vaccine.

```
p2 +
  geom_vline(xintercept = 1963, col = "gray") +
  geom_vline(xintercept = 1955, col = "gray")
```



6 Auto-generated Plot (4.3): GGPairs

```
library("GGally")

ri_measles_pre = diseases %>%
  filter(state == "Rhode Island", disease == "Measles", year < 1963) %>%
  select(year, weeks_reporting, count, population, rate)

ggpairs(ri_measles_pre)
```



7 Hypothesis Testing Examples

7.1 Example 1: Two-sample t-test (RI vs non-RI measles pre-1963)

Let μ_{RI} be the mean rate of measles in RI prior to 1963 and $\mu_{US\setminus RI}$ be the mean rate of measles in the US excluding RI prior to 1963.

H_0: $\mu_{RI} = \mu_{US\setminus RI}$ H_A: $\mu_{RI} \neq \mu_{US\setminus RI}$

```
rates_ri = diseases %>%
  filter(state == "Rhode Island", disease == "Measles", year <= 1963) %>%
  pull(rate)

rates_nonri = diseases %>%
  filter(state != "Rhode Island", disease == "Measles", year <= 1963) %>%
  pull(rate)

t.test(rates_ri, rates_nonri)
```

```
    Welch Two Sample t-test
```

```
data:  rates_ri and rates_nonri
t = 0.82434, df = 36.222, p-value = 0.4151
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -78.98143 187.19385
sample estimates:
mean of x mean of y
 421.6372  367.5310
```

7.2 Example 2: Chi-square test of independence (RI vs non-RI and high_rate)

We test whether having a high rate of measles (rate > 100) is related to being in RI vs not. H_0: independent H_A: related

```r
diseases_measles = diseases %>%
  filter(year <= 1963, disease == "Measles") %>%
  mutate(ri = (state == "Rhode Island"))

chisq.test(table(diseases_measles$ri, diseases_measles$high_rate), correct = FALSE)
```

```
    Pearson's Chi-squared test

data:  table(diseases_measles$ri, diseases_measles$high_rate)
X-squared = 1.1585, df = 1, p-value = 0.2818
```

7.3 Example 3: Vaccine effectiveness (measles pre vs post 1963)

Here, we test the vaccine effectiveness claim: Do vaccines seem effective? H_0: mean rate before 1963 equals mean rate after 1963 H_A: means differ

```r
measles_pre = diseases %>%
  filter(disease == "Measles", year < 1963) %>%
  pull(rate)

measles_post = diseases %>%
  filter(disease == "Measles", year >= 1963) %>%
  pull(rate)

t.test(measles_pre, measles_post)
```

```
    Welch Two Sample t-test

data:  measles_pre and measles_post
t = 37.179, df = 2001.5, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 318.3749 353.8333
sample estimates:
mean of x mean of y
371.73167  35.62758
```

8 Writing up your EDA

Whenever you analyze a dataset, it is useful to write up your findings in a clear report.

We analyze a dataset of infectious disease records in the US. The dataset used comes from the Tycho project and can be downloaded using the dslabs package in R. Each row corresponds to a state's yearly count record for a contagious disease, and the variables are **disease, state, year, weeks_reporting, count, and population**. Weeks_reporting corresponds to the total number of weeks that disease counts were recorded that year and population corresponds to the state's population for that year (interpolated for non-census years). In our analysis, we will focus on incidents of contagious diseases in RI. We want to understand how the prevalence of diseases changed over time and how these trends relate to the rest of the US.

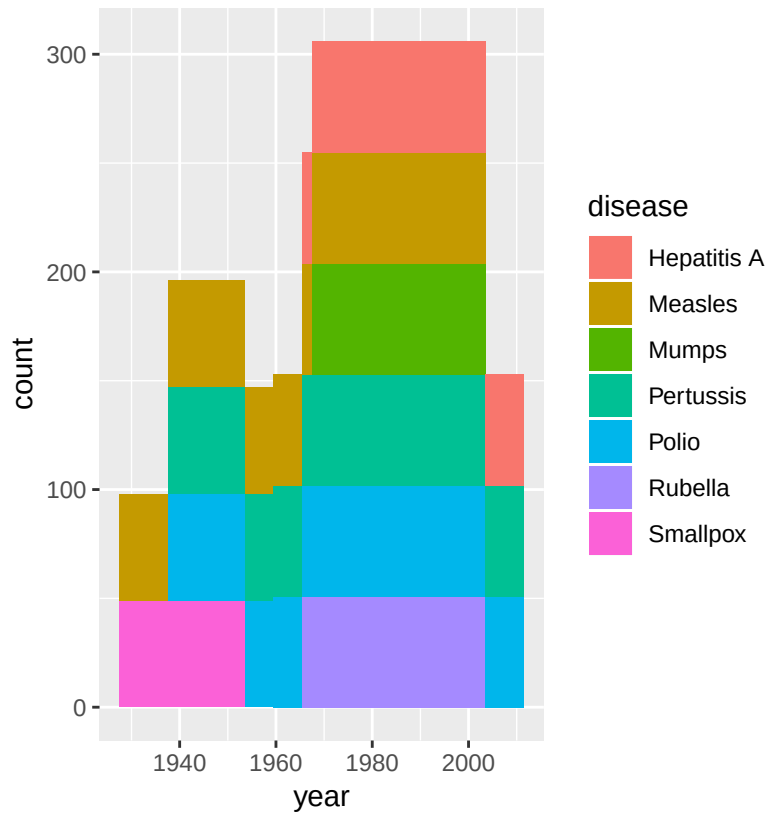
We first checked for any missing data and found that only around 1% of the data was missing. This corresponded to entries for which the weeks_reporting was missing. We decided to omit this data in our analysis.

We next looked at the individual variables. Every state is evenly represented within the dataset with 370 entries each and the year range is from 1928-2011. However, we do not have a record for each disease in each year in each state. In particular, the proportion of records for each disease is given below with measles being the most common and smallpox being the least.

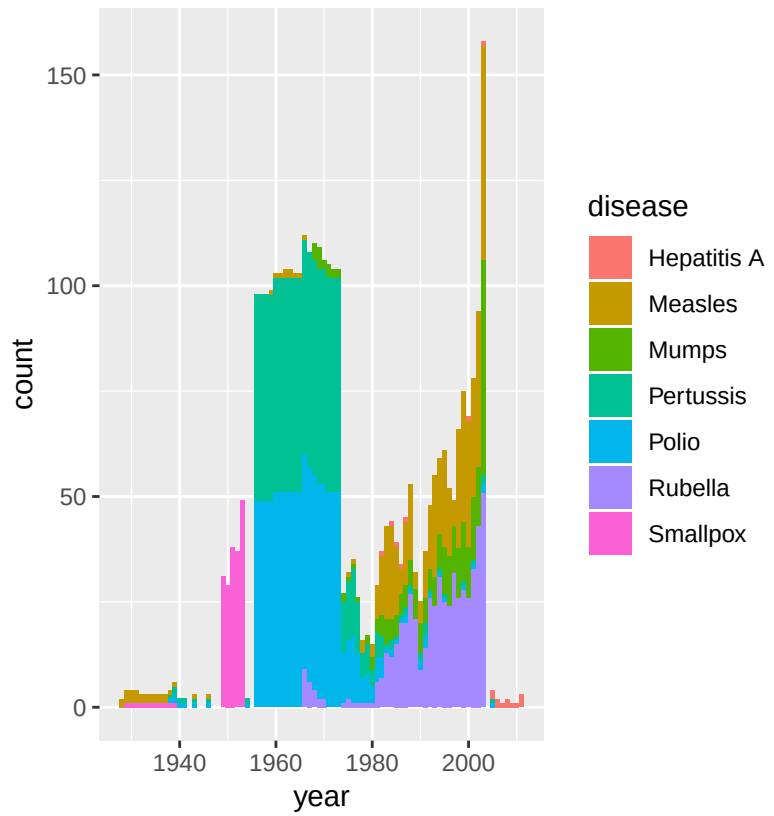
```r
disease_summary = diseases %>%
  count(disease) %>%
  mutate(proportion = n / sum(n))

knitr::kable(disease_summary, digits = 3)
```

disease	n	proportion
Hepatitis A	2327	0.164
Measles	3319	0.233
Mumps	1576	0.111
Pertussis	2709	0.190
Polio	1844	0.130
Rubella	1374	0.097
Smallpox	1079	0.076

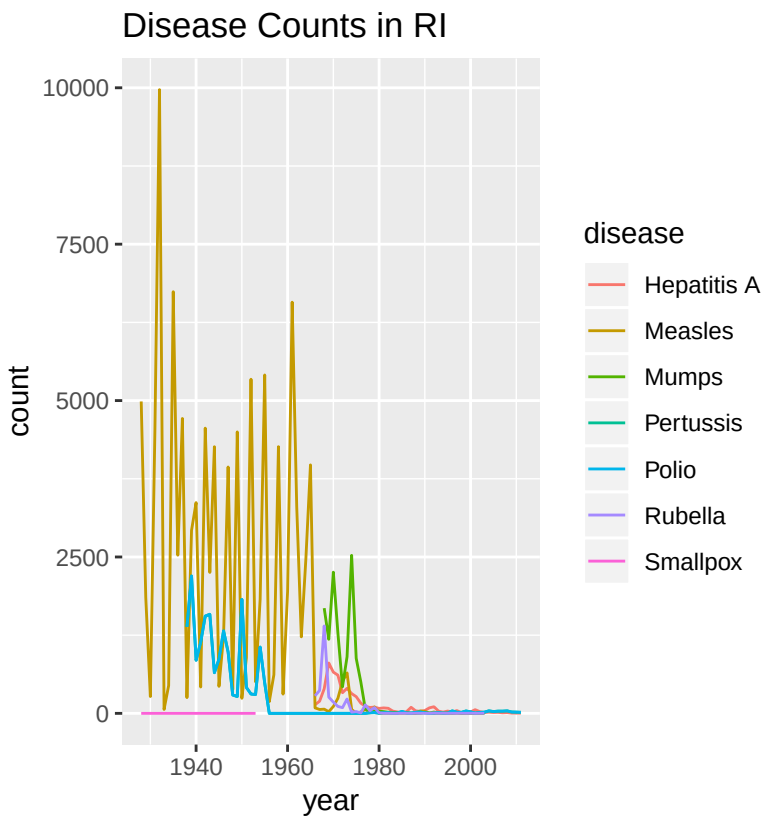
```
knitr::include_graphics("year.dist.pdf")
```

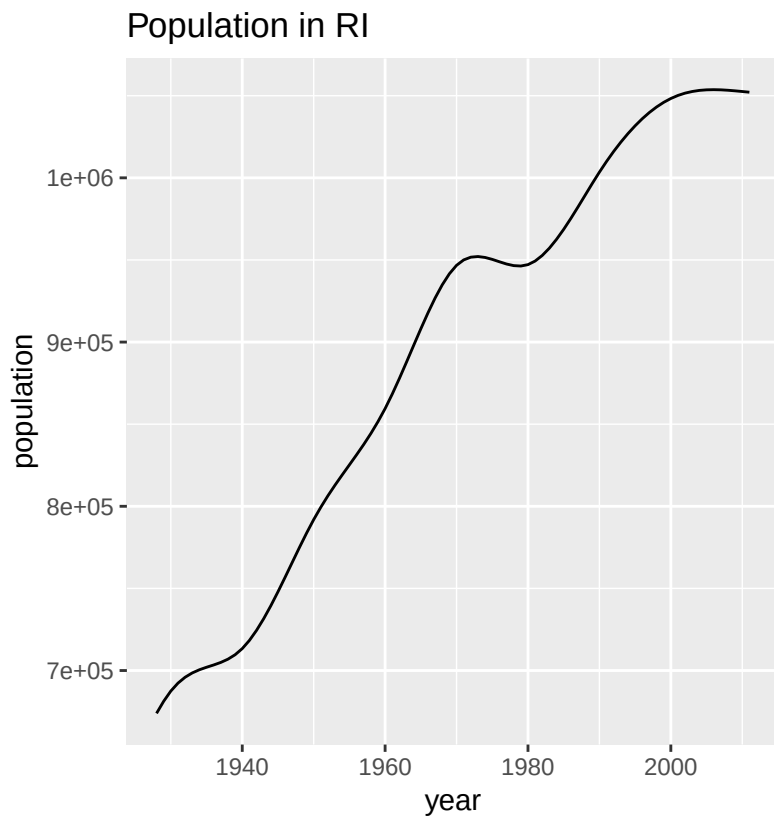



```
knitr::include_graphics("missing.weeks.pdf")
```

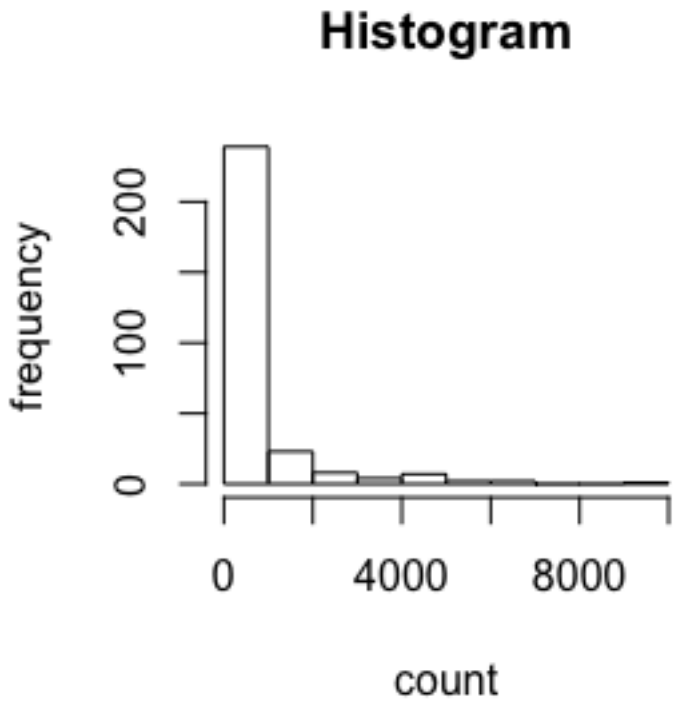


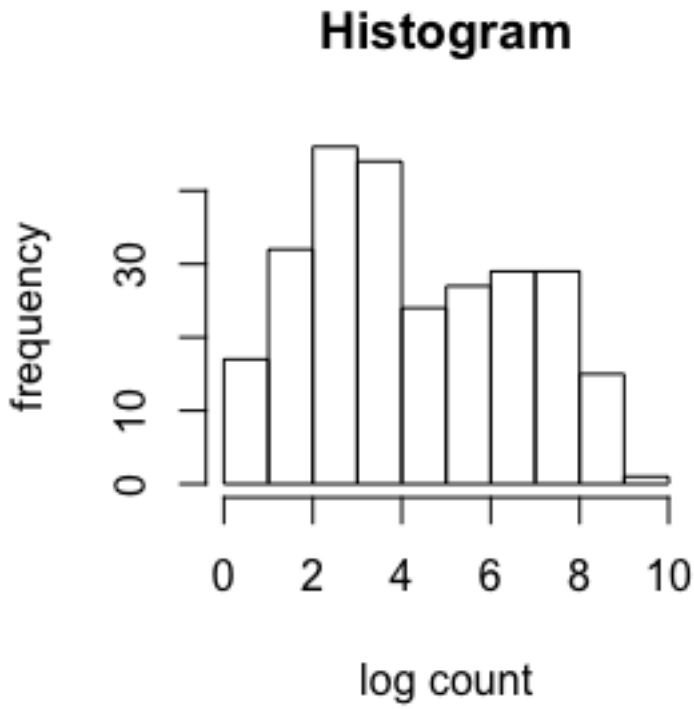

```
knitr::include_graphics(c("counts.ri.pdf", "population.ri.pdf"))
```


Disease Counts in RI
10000
7500
5000
2500
0
count
1940
1960
1980
2000
year
disease
Hepatitis A
Measles
Mumps
Pertussis
Polio
Rubella
Smallpox

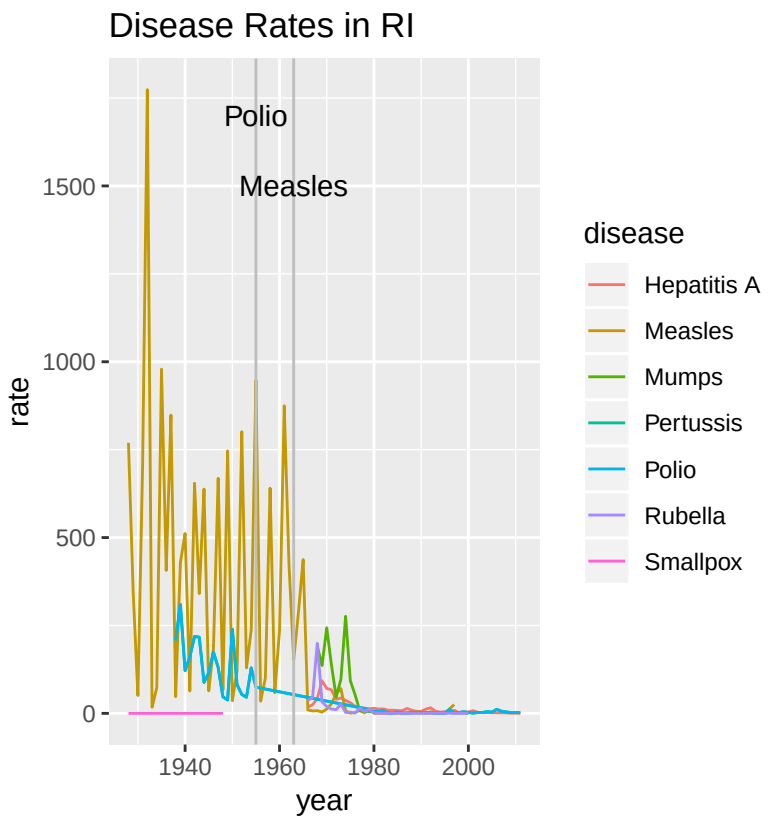


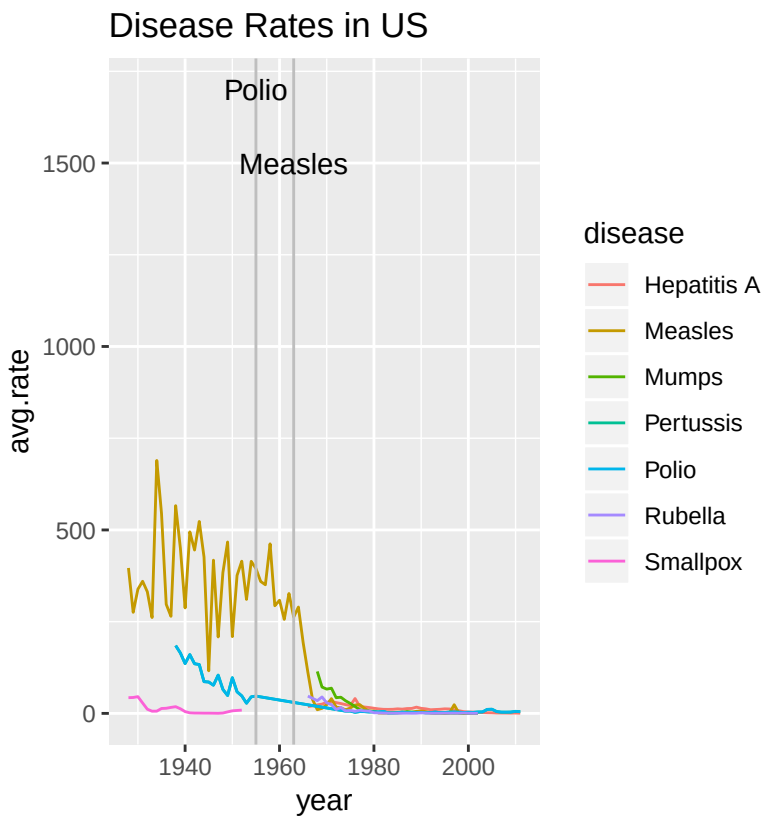

```
knitr::include_graphics(c("hist.count.png", "log.count.png"))
```


Histogram
frequency
0
100
200
0
4000
8000
count




```
knitr::include_graphics(c("rates.ri.pdf", "rates.us.pdf"))
```


Disease Rates in RI
Polio
Measles
1500
1000
500
0
rate
1940
1960
1980
2000
year
disease
Hepatitis A
Measles
Mumps
Pertussis
Polio
Rubella
Smallpox



While this dataset gave evidence for the effectiveness of vaccines, in future work, it would be interesting to focus on any noticeable resurgence in diseases from the anti-vaccine movement. To do so, we would likely need to collect the most recent disease counts. Also, it would be interesting to look at the characteristics of more recent cases (both individual characteristics such as age and state characteristics).